Python Basics (Foundation for Data Analytics)

Variables & Basic Operations

1. Take 5 numbers as input and calculate their average.
2. Write a program to check whether a number is even or odd.
3. Convert temperature from Celsius to Fahrenheit.
4. Write a program to print the square and cube of a number.

Strings (Very Important for Data Cleaning)

5. Take a sentence as input and:
    - Count total characters
    - Count total words
6. Count the number of vowels in a string.
7. Check whether an email is valid (must contain "@" and ".").
8. Reverse a string.
9. Convert all words in a sentence to uppercase.
10. Remove extra spaces from a string.

Lists (Core for Data Handling)

11. Create a list of 10 numbers and find:

    - Maximum
    - Minimum
    - Sum
    - Average

12. Remove duplicate values from a list.
13. Sort a list in ascending and descending order.
14. Store negative numbers from a list into a new list.
15. Extract only even numbers from a list.

Loops (Data Processing Logic)

16. Print numbers between 1–100 that are divisible by 3.
17. Calculate factorial of a number.
18. Print Fibonacci series up to n terms.
19. Count how many numbers in a list are greater than 50.
20. Store squares of all elements of a list in a new list.

Data-Oriented Logic Building

Dictionaries (Very Important – Future Pandas Concept)

21. Create a dictionary of student marks and find the student with highest marks.
22. Create a word frequency counter from a sentence.
23. Sort a dictionary by keys.
24. Find the sum of all values in a dictionary.

Functions (Important for Reusability)

25. Create a function that returns the average of a list.
26. Create a function to check whether a number is prime.
27. Create a function to detect outliers in a list
    (Hint: value > mean + 2 × standard deviation).

Mini Data Problems (Real Analytics Thinking)

28. Given:

```
sales = [1200, 1500, 1100, 1800, 900, 2000]
```

- Total sales
- Average sales
- Best sale day
- Worst sale day

29. Given:

```
marks = [45, 67, 89, 34, 90, 55]
```

- Count Pass/Fail (pass ≥ 40)
- Assign grades (A, B, C)

30. Detect missing values in a list

Level 3: Analytics Preparation

31. Generate 50 random numbers and calculate:

- Mean
- Median
- Mode
  (Without using pandas)

32. Given:

```
data = [
    ["Rahul", 25, 50000],
    ["Amit", 30, 60000],
    ["Neha", 22, 45000]
]
```

- Calculate average salary
- Print names of people with age > 24

33. Perform normalization on a list using:

    (value−min)/(max−min)(value - min) / (max - min)(value−min)/(max−min)

34. Write a program to extract the top 3 largest values from a list.

Bonus Mini Project (Highly Recommended)

35. Student Result Analysis System

Build a program that:

- Takes multiple students' data
- Calculates total and average
- Finds topper
- Counts failed students
- Shows grade distribution